

Run 05 | OME

India

IV only

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0

Sample 27,555 respondents **Coverage** 1 country; 2015–2022

Positive cases 2,167 (7.86%) **Case time** 35s

Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9851). Highest F1: Random Forest (0.7719). The all-negative accuracy baseline is 92.14% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9851	0.8136	95.93	69.03	87.53	0.7719	<1s
H2O GBM	0.9849	0.8050	95.93	69.24	86.84	0.7705	2s
Logistic Regression	0.9848	0.8052	95.70	66.23	92.38	0.7715	<1s
Lasso	0.9847	0.8038	95.79	70.14	80.83	0.7511	8s
XGBoost	0.9847	0.8013	95.90	69.49	85.22	0.7656	<1s
CatBoost	0.9846	0.8071	95.88	71.46	79.21	0.7514	3s
Ridge	0.9846	0.8054	95.64	65.90	92.38	0.7692	7s
Elastic Net	0.9844	0.8017	95.90	70.82	81.29	0.7570	9s
Decision Tree	0.9830	0.7542	95.54	67.88	81.99	0.7427	<1s
Naive Bayes	0.9065	0.5726	92.98	54.06	70.67	0.6126	3s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	893.91	49.95
2	Established business owner	373.76	20.89
3	Business angel	82.87	4.63
4	Knowing entrepreneurs	69.51	3.88
5	Business exit	42.15	2.36
6	Self-efficacy	27.40	1.53
7	Business exit (missing indicator)	24.34	1.36
8	Social enterprise	23.94	1.34

Share is the percentage of total forest impurity importance across all 25 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 05_obj1_india_iv_OME. Scores and importance are taken directly from saved CSV outputs.